Equity Performance Project

Analysis of Investor Personas and Index Tracking

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Motivation

- Modern Portfolio Theory provides a framework for risk–return tradeoffs.
- We evaluate a 10-stock equity portfolio under two investor personas.
- Risk metrics: Sharpe Ratio, VaR, CVaR, volatility decomposition.

Portfolio Description

10-Stock Portfolio:

- Technology: MSFT, AAPL, IBM
- Dividend Aristocrats: XOM, WTM, KO, JNJ
- High Dividend Yield: VZ, KHC, USB

Investor Personas

Buy & Hold (BH):

- Long-term investor, single wealth trajectory.
- P&L: $\text{Value}_E - \text{Value}_S$

Hedge Fund (HF):

- Daily mark-to-market.
- P&L series: $\text{Value}_t - \text{Value}_{t-1}$

Risk and Performance Metrics

- Sharpe Ratio
- Value at Risk (Historical Simulation)
- Conditional Value at Risk
- Volatility estimation and rolling windows

Weighting Methodologies

- Equal Weighting
- Market Capitalization Weighting
- Logarithmic Market Cap Weighting

Index Tracking Optimization

Objective: Minimize Tracking Error Variance:

$$\min_{w} \mathbb{E}[(R_T - R_R)^2]$$

Constraints:

- Full investment: $\sum w_i = 1$
- Long-only: $w_i \geq 0$
- Cardinality: $\|w\|_0 \leq k$
- Buy-in threshold: $w_i \in \{0\} \cup [l, 1]$

Summary of Findings

- Market-cap weighting delivered high returns but extreme concentration.
- Equal weighting provided stable diversification but lower risk-adjusted returns.
- Log-market-cap weighting offered a balanced alternative.
- Index tracking achieved low tracking error under practical constraints.

Conclusions

- Investor objectives strongly influence optimal portfolio design.
- Weighting choices drive concentration, turnover, and performance.
- Optimization frameworks enable realistic index replication.

Thank you!